

Microbiome Diversity in Soils of the Agro-Ecological Zones of Bangladesh

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Abstract

Diversity and composition of soil bacteriome across various agro-ecological zones (AEZs) in Bangladesh were unveiled using 16S rRNA gene sequencing. The mapping of bacterial communities in each AEZ has laid a foundation for detailed mapping and engineering of soil microbiome in deltaic Bangladesh soils for mitigating the impact of climate change and promoting sustainable agriculture.

Keywords: Agro-ecological zones, Soil microbiome, Soil bacteriome, Microbial diversity, Bangladesh soil

Announcement

Soil microbiome plays an important role in maintaining soil health, plant productivity, and soil ecosystem services. Bangladesh heavily relies on intensive agriculture for food security, rural development, and poverty alleviation. Despite developing soil maps for all AEZs based on physico-chemical properties, the microbiome — key to nutrient cycling, soil productivity, and greenhouse gas dynamics — remains largely overlooked. Better understanding of microbial diversity across 30 AEZs in Bangladesh is critical for sustainable agriculture and climate change mitigation.

Soil samples were collected from 14 AEZs during March–August 2024. Sampling sites spanned coordinates 22.17–25.879°N and 88.25–92.18°E, characterized by a tropical wet and dry climate. Composite soil samples were prepared by pooling three samples from each zone (depth 0–5 cm). DNA was extracted using the Qiagen DNeasy PowerSoil Kit. Amplicon libraries targeted V3–V4 regions with 341F/806R primers and were sequenced on the Illumina MiSeq platform (2×300 bp). Data processing used QIIME 2 (2023.2.0) and SILVA v.138 (≥98% similarity OTU clustering).

From 14 composite samples, 7,199,364 raw reads were obtained (avg 514,240/sample). After quality filtering, 343,166 reads (4.77%) were assigned to 544 OTUs spanning 16 phyla, 45 classes, 77 orders, 151 families, and 341 genera. *Gemmata* (12.11%) was the top abundant genus, followed by *Planctomyces* (8.83%), *Nocardioides* (6.11%), *Anaerolinea* (5.28%), and others.

Table 1: Sample Information and Sequencing Statistics

Sample ID	Location (abbrev.)	Coordinate	pH	SOM	Texture	Raw Reads	Mapped Reads	OTUs	Genera	SRA
AEZ-2	Nilphamari et al.	25.879°N, 88.257°E	Acidic (4.6)	Low (1.21)	Loam	446,896	26,303	363	227	SRR31219345
AEZ-4	Bogra et al.	24.671°N, 89.546°E	Acidic (5.1)	Medium (2.82)	Silt loam	810,184	49,730	444	290	SRR31219339
AEZ-5	Naogaon et al.	24.529°N, 89.175°E	Acidic (5.3)	Medium (2.69)	Silt loam	487,106	33,056	364	239	SRR31219334
AEZ-10	Ganges/Meghna channels et al.	23.746°N, 89.769°E	Alkaline (7.1)	Low (1.88)	Silt clay loam	367,152	19,691	360	237	SRR31219333
AEZ-12	Pabna et al.	23.572°N, 89.825°E	Acidic (6.9)	Medium (2.08)	Clay	574,064	20,790	305	210	SRR31219332
AEZ-13	Barisal et al.	22.698°N, 89.518°E	Alkaline (7.2)	Low (1.08)	Silt loam	530,834	20,114	339	227	SRR31219344
AEZ-16	Kishoreganj et al.	23.577°N, 90.788°E	Acidic (5.8)	Low (1.48)	Loam	508,112	29,252	398	262	SRR31219343
AEZ-17	Chandpur et al.	23.120°N, 90.758°E	Acidic (5.7)	Medium (2.15)	Silt loam	255,788	15,144	304	202	SRR31219342
AEZ-19	Kishoreganj et al.	23.185°N, 89.810°E	Acidic (6.6)	Medium (2.89)	Clay loam	341,424	16,440	359	235	SRR31219341
AEZ-25	Dinajpur et al.	24.656°N, 89.323°E	Acidic (5.7)	Low (1.82)	Clay loam	528,176	27,085	374	252	SRR31219340

Sample ID	Location (abbrev.)	Coordinate	pH	SOM	Texture	Raw Reads	Mapped Reads	OTUs	Genera	SRA
AEZ-26	Rajshahi et al.	24.849°N, 88.440°E	Acidic (6.0)	Low (1.86)	Clay loam	570,734	35,620	445	280	SRR31219338
AEZ-27	Dinajpur et al.	25.616°N, 89.190°E	Acidic (5.3)	Low (1.08)	Silt loam	342,928	16,257	305	208	SRR31219337
AEZ-28	Dhaka et al.	24.665°N, 90.030°E	Acidic (3.4)	Medium (2.29)	Sandy loam	819,726	14,447	208	125	SRR31219336
AEZ-29	Khagrachhari et al.	22.170°N, 92.189°E	Acidic (4.1)	Medium (2.35)	Loam	616,240	19,237	226	149	SRR31219335

SOM = Soil Organic Matter. OTUs grouped at ≥98% similarity against SILVA v.138.

Data Availability

16S rRNA gene amplicon sequencing data are available at the NCBI Sequence Read Archive (SRA) under BioProject accession number PRJNA1181648.

Selected References

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